A wonderful venue for geometrodynamics is the collision of two black holes. When they collide, the holes set space and time into wild gyrations. Our SXS simulations have now reached maturity, and are beginning to reveal relativity's predictions (Figure 16.9). LIGO and its partners will observe the gravitational waves from colliding black holes within the next few years, and test our simulations' predictions. It's a wonderful era for probing geometrodynamics!

Gravitational Waves from the Big Bang

In 1975 Leonid Grishchuk, a dear Russian friend of mine, made a startling prediction: A rich plethora of gravitational waves was produced in the big bang, he predicted, by a previously unknown mechanism: Quantum fluctuations of gravity coming off the big bang were amplified enormously, he told us, by the universe's initial expansion; and when amplified, they became primordial gravitational waves. If discovered, these gravitational waves could bring us a glimpse of our universe's birth.

In subsequent years, as our understanding of the big bang matured, it became evident that the waves would be strongest at wavelengths nearly as large as the visible universe itself, billions of light-years' wavelength, and would likely be too weak for detection at LIGO's far shorter wavelengths, hundreds and thousands of kilometers.

In the early 1990s several cosmologists realized that these billion-light-years-long *gravitational* waves should have placed a unique imprint on *electromagnetic* waves that fill the universe, the so-called cosmic microwave background or CMB. A holy grail quickly emerged: search for that CMB imprint, from it infer the properties of the primordial gravitational waves that produced the imprint, and thereby explore the birth of the universe.